

Quick Check

Lesson 4: Food Tests Practical Part 1 — Detecting Starch and Reducing Sugars in Kenyan Foods

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.3: Nutrition in Animals | Lesson 4

Before testing ugali, a learner recalls the enzymes from the last lesson. Which food substance does salivary amylase act on?

A Starch

B Water

C Fat

D Protein

A group tests a glucose solution as the control. What results should they expect?

- A** Iodine: blue-black; Benedict: stays blue
- B** Iodine: brown-orange; Benedict: stays blue
- C** Iodine: blue-black; Benedict: orange-red after heating
- D** Iodine: brown-orange; Benedict: orange-red after heating

Raw maize flour turns blue-black when two drops of iodine are added. What does this result show?

A Starch is present in the maize flour even before it is chewed

B Starch is absent from the maize flour

C Reducing sugars are present in the maize flour

D Salivary amylase has already digested the starch

During the Benedict test, a test tube is heated in a hot water bath for five minutes. What is the safe way to handle the tube?

- A** Leave the tube in the boiling water bath and lift it out with your hand when the colour changes
- B** Hold it with bare fingers so you can feel when it is hot enough
- C** Use a test-tube holder and keep the tube at an angle, pointing away from your face and others
- D** Hold it upright with a test-tube holder and look down into it

A ripe banana gives an orange-red result with Benedict reagent, but an unripe banana gives a much weaker result. What best explains this?

- A** Benedict reagent detects starch, which is present in ripe bananas
- B** Ripe bananas never contained starch at any stage
- C** Iodine solution has changed the banana into sugar
- D** During ripening, starch in the banana is converted into reducing sugars

Cooked ugali gives a blue-black colour with iodine. Which is the best evidence statement to write in your notebook?

- A** Ugali contains no carbohydrate, because it does not taste sweet
- B** Ugali contains starch, which is the substrate for lipase
- C** Ugali contains starch, which is a substrate for salivary amylase and pancreatic amylase
- D** Ugali contains reducing sugars, because starch and sugar are the same substance

A learner draws a two-box diagram for ugali: Box 1 is MOUTH and Box 2 is SMALL INTESTINE. Which pair of labels is correct?

- A** Mouth: no digestion of starch happens; small intestine: salivary amylase digests all the starch
- B** Mouth: salivary amylase breaks starch into maltose; small intestine: lipase digests the remaining starch
- C** Mouth: pepsin breaks starch into maltose; small intestine: salivary amylase digests the remaining starch
- D** Mouth: salivary amylase starts breaking starch into maltose; small intestine: pancreatic amylase digests the remaining starch

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).